

Lecture 11

Conjugate priors: Part 3

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(Edited from the lecture notes by Brian J. Reich, NC State)



Conjugate priors: Recap

- ▶ A prior is called **conjugate** if the posterior belongs to the same parametric family as that of the prior.
- ▶ We have seen that in case of a binomial or negative binomial likelihood and a beta prior, the posterior is also a beta distribution.
- ▶ In case of a multinomial likelihood and a Dirichlet prior, the posterior distribution is also Dirichlet.
- ▶ In case of a Poisson or exponential or gamma (with known shape parameter) likelihood and a gamma prior, the posterior distribution is also gamma.

Gaussian models (univariate)

- ▶ The final (and the most important) distribution we will discuss is the Gaussian (normal) distribution, and we denote it by $Y \sim \text{Normal}(\mu, \sigma^2)$.
 - ▶ Domain: $Y \in (-\infty, \infty)$.
 - ▶ PDF: $f(y) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{1}{2} \left(\frac{y-\mu}{\sigma} \right)^2 \right]$.
 - ▶ Mean: $E(Y|\mu, \sigma^2) = \mu$.
 - ▶ Variance: $V(Y|\mu, \sigma^2) = \sigma^2$.
- ▶ Today we will discuss:
 - ▶ Estimating the mean assuming the variance is known.
 - ▶ Estimating the variance assuming the mean is known.
- ▶ For estimating mean and variance both, we will use **MCMC**!

Estimating a normal mean with known variance - Likelihood

- ▶ We assume the data consist of n independent and identically distributed observations $Y_1, \dots, Y_n$. Let $\mathbf{Y} = (Y_1, \dots, Y_n)'$.
- ▶ Each Y_i is Gaussian (given μ),

$$Y_i | \mu \sim \text{Normal}(\mu, \sigma^2),$$

where σ is known.

- ▶ The likelihood is then

$$\prod_{i=1}^n f(y_i | \mu) = \left(\frac{1}{\sqrt{2\pi}\sigma} \right)^n \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2 \right].$$

Bayesian analysis - Prior

- ▶ The parameter μ is continuous over the entire real line, therefore a natural prior is

$$\mu \sim \text{Normal}(\theta, \eta^2).$$

- ▶ The prior mean θ is the best guess before we observe data.
- ▶ The math is slightly more interpretable if we set $\eta^2 = \frac{\sigma^2}{m}$.
- ▶ As we will see, the prior variance controls the strength of the prior via $m > 0$.

Derivation of the posterior

- The posterior is

$$\begin{aligned} p(\mu|\mathbf{Y}) &\propto \exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^n(Y_i - \mu)^2\right] \times \exp\left[-\frac{m}{2\sigma^2}(\mu - \theta)^2\right] \\ &\propto \exp\left[-\frac{n\mu^2}{2\sigma^2}\right] \times \exp\left[\frac{\mu\sum_{i=1}^n Y_i}{\sigma^2}\right] \times \exp\left[-\frac{m\mu^2}{2\sigma^2}\right] \times \exp\left[\frac{m\mu\theta}{\sigma^2}\right] \\ &= \exp\left[-\frac{1}{2}\mu^2\frac{n+m}{\sigma^2}\right] \times \exp\left[\mu\frac{(\sum_{i=1}^n Y_i + m\theta)}{\sigma^2}\right]. \end{aligned}$$

By matching the terms with those in the density of a normal distribution, we have

$$\mu|\mathbf{Y} \sim \text{Normal}\left(\frac{\sum_{i=1}^n Y_i + m\theta}{n+m}, \frac{\sigma^2}{n+m}\right).$$

Shrinkage

- ▶ The posterior mean is

$$\hat{\mu}_B = E(\mu|\mathbf{Y}) = \frac{\sum_{i=1}^n Y_i + m\theta}{n + m} = w\bar{Y} + (1 - w)\theta,$$

where $w = n/(n + m)$.

- ▶ Therefore, if m is small then $\hat{\mu}_B \approx \bar{Y}$, and if m is large $\hat{\mu}_B \approx \theta$.
- ▶ If no prior information is available, take m to be small and thus the prior is non-informative.
- ▶ Small m gives large prior variance (relative to σ).

Shrinkage

- ▶ The posterior variance is

$$V(\mu|\mathbf{Y}) = \frac{\sigma^2}{n + m}.$$

- ▶ The sampling variance of $\bar{Y}$ is $\frac{\sigma^2}{n}$.
- ▶ Therefore, we can loosely interpret m as the “prior number of observations”.
- ▶ If m is small then $V(\mu|\mathbf{Y}) \approx V(\bar{Y})$, and if m is large $V(\mu|\mathbf{Y}) \approx V(\mu)$.

Estimating a normal variance with known mean - Likelihood

- ▶ We assume the data consist of n independent and identically distributed observations $Y_1, \dots, Y_n$. Let $\mathbf{Y} = (Y_1, \dots, Y_n)'$.
- ▶ Each Y_i is Gaussian (given σ^2),

$$Y_i | \sigma^2 \sim \text{Normal}(\mu, \sigma^2),$$

where μ is known.

- ▶ The likelihood is then

$$\prod_{i=1}^n f(y_i | \sigma^2) = \left(\frac{1}{\sqrt{2\pi}\sigma} \right)^n \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mu)^2 \right].$$

Bayesian analysis - Prior

- ▶ The parameter σ^2 is continuous over $(0, \infty)$, therefore a natural prior is $\sigma^2 \sim \text{Gamma}(a, b)$.
- ▶ However, the math is easier if we pick a gamma prior for the inverse variance (precision) $1/\sigma^2$.
- ▶ If $1/\sigma^2 \sim \text{Gamma}(a, b)$, then $\sigma^2 \sim \text{Inverse-Gamma}(a, b)$.
- ▶ This is the definition of the inverse-gamma distribution (Lecture 1).
- ▶ The inverse-gamma prior for σ^2 has PDF

$$f(\sigma^2) = \frac{b^a}{\Gamma(a)} (\sigma^2)^{-a-1} \exp(-b/\sigma^2).$$

Derivation of the posterior

- ▶ The posterior is

$$\begin{aligned} p(\sigma^2 | \mathbf{Y}) &\propto (\sigma^2)^{-n/2} \exp \left[-\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \mu)^2 \right] \times (\sigma^2)^{-a-1} \exp(-b/\sigma^2) \\ &= (\sigma^2)^{-(n/2+a)-1} \exp \left[-\frac{1}{\sigma^2} \left(\frac{1}{2} \sum_{i=1}^n (Y_i - \mu)^2 + b \right) \right]. \end{aligned}$$

- ▶ The posterior is

$$\sigma^2 | \mathbf{Y} \sim \text{Inverse-Gamma}(n/2 + a, SSE/2 + b),$$

where $SSE = \sum_{i=1}^n (Y_i - \mu)^2$.

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- ▶ The mean of an Inverse-Gamma(a, b) distribution only exists if $a > 1$.
- ▶ The prior mean (if it exists) is $b/(a - 1)$.

- ▶ The posterior mean is

$$\frac{SSE + 2b}{n + 2a - 2}$$

- ▶ It is common to take a and b to be small to give a non-informative prior.
- ▶ For large n and comparatively negligible a and b , the posterior mean is approximately same as the sample variance SSE/n .

Conjugate prior for a normal precision

- ▶ The precision is the inverse variance, $\tau = 1/\sigma^2$.
- ▶ If Y_i have mean μ and precision τ , the likelihood is proportional to

$$\prod_{i=1}^n f(y_i|\tau) \propto \tau^{n/2} \exp \left[-\frac{\tau}{2} \sum_{i=1}^n (y_i - \mu)^2 \right]$$

- ▶ If $\tau \sim \text{Gamma}(a, b)$, then

$$\tau|Y \sim \text{Gamma}(n/2 + a, SSE/2 + b).$$

- ▶ This is the exact same analysis as the inverse-gamma prior for the variance.

Gaussian models (multivariate)

- ▶ The final multivariate distribution we will discuss is the multivariate normal distribution, and we denote the P -variate case by $\mathbf{Y} \sim \text{Normal}_P(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.
 - ▶ Domain: $\mathbf{Y} \in \mathbb{R}^P = (-\infty, \infty)^P$.
 - ▶ PDF: $f(\mathbf{y}) = |2\pi\boldsymbol{\Sigma}|^{-1/2} \exp \left[-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu}) \right]$.
 - ▶ Mean vector: $E(\mathbf{Y}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \boldsymbol{\mu}$.
 - ▶ Covariance matrix: $\text{Cov}(\mathbf{Y}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \boldsymbol{\Sigma}$. Here, $\boldsymbol{\Sigma}$ is a positive definite (PD) matrix.
- ▶ Today we will discuss:
 - ▶ Estimating the mean vector assuming the covariance matrix is known.
 - ▶ Estimating the covariance matrix assuming the mean vector is known.
- ▶ For estimating mean vector and covariance matrix both, we will use **MCMC**!

Estimating a normal mean vector with known covariance - Likelihood

- ▶ We assume the data consist of n independent and identically distributed observation vectors $\mathbf{Y}_1, \dots, \mathbf{Y}_n$. Let $\mathcal{Y} = \{\mathbf{Y}_1, \dots, \mathbf{Y}_n\}$.
- ▶ Each $\mathbf{Y}_i$ is Gaussian (given $\boldsymbol{\mu}$),

$$\mathbf{Y}_i | \boldsymbol{\mu} \sim \text{Normal}_P(\boldsymbol{\mu}, \boldsymbol{\Sigma}),$$

where $\boldsymbol{\Sigma}$ is known.

- ▶ The likelihood is then

$$\prod_{i=1}^n f(\mathbf{y}_i | \boldsymbol{\mu}) = |2\pi\boldsymbol{\Sigma}|^{-n/2} \exp \left[-\frac{1}{2} \sum_{i=1}^n (\mathbf{y}_i - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{y}_i - \boldsymbol{\mu}) \right].$$

Bayesian analysis - Prior

- ▶ The parameter vector μ is continuous over $\mathbb{R}^P$, therefore a natural prior is

$$\mu \sim \text{Normal}_P(\mu_0, \Sigma_0).$$

- ▶ The prior mean μ_0 is the best guess before we observe data.
- ▶ The math is slightly more interpretable if we set $\Sigma_0 = \frac{1}{m}\Sigma$.
- ▶ As we will see, the prior variance controls the strength of the prior via $m > 0$.

Derivation of the posterior

- The posterior is

$$\begin{aligned} p(\boldsymbol{\mu}|\mathcal{Y}) &\propto \exp \left[-\frac{1}{2} \sum_{i=1}^n (\mathbf{Y}_i - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{Y}_i - \boldsymbol{\mu}) \right] \times \exp \left[-\frac{m}{2} (\boldsymbol{\mu} - \boldsymbol{\mu}_0)' \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu} - \boldsymbol{\mu}_0) \right] \\ &\propto \exp \left[-\frac{n+m}{2} \boldsymbol{\mu}' \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} \right] \times \exp \left[\boldsymbol{\mu}' \boldsymbol{\Sigma}^{-1} \left(\sum_{i=1}^n \mathbf{Y}_i + m \boldsymbol{\mu}_0 \right) \right]. \end{aligned}$$

By matching the terms with those in the density of a multivariate normal distribution, we have

$$\boldsymbol{\mu}|\mathcal{Y} \sim \text{Normal}_P \left(\frac{\sum_{i=1}^n \mathbf{Y}_i + m \boldsymbol{\mu}_0}{n+m}, \frac{1}{n+m} \boldsymbol{\Sigma} \right).$$

Shrinkage

- ▶ The posterior mean is

$$\hat{\mu}_B = E(\mu|\mathcal{Y}) = \frac{\sum_{i=1}^n \mathbf{Y}_i + m\mu_0}{n+m} = w\bar{\mathbf{Y}} + (1-w)\mu_0,$$

where $w = n/(n+m)$ and $\bar{\mathbf{Y}} = \frac{1}{n} \sum_{i=1}^n \mathbf{Y}_i$.

- ▶ Therefore, if m is small then $\hat{\mu}_B \approx \bar{\mathbf{Y}}$, and if m is large $\hat{\mu}_B \approx \mu_0$.
- ▶ If no prior information is available, take m to be small and thus the prior is non-informative.
- ▶ Small m gives large prior variances and covariances (relative to the components of Σ).

Shrinkage

- ▶ The posterior covariance matrix is

$$\text{Cov}(\boldsymbol{\mu}|\mathcal{Y}) = \frac{1}{n+m}\boldsymbol{\Sigma}.$$

- ▶ The sampling variance of $\bar{\mathbf{Y}}$ is $\frac{1}{n}\boldsymbol{\Sigma}$.
- ▶ Therefore, we can loosely interpret m as the “prior number of observations”.
- ▶ If m is small then $\text{Cov}(\boldsymbol{\mu}|\mathcal{Y}) \approx \text{Cov}(\bar{\mathbf{Y}})$, and if m is large $\text{Cov}(\boldsymbol{\mu}|\mathcal{Y}) \approx \text{Cov}(\boldsymbol{\mu})$.

Estimating a normal covariance matrix with known mean - Likelihood

- ▶ We assume the data consist of n independent and identically distributed observation vectors $\mathbf{Y}_1, \dots, \mathbf{Y}_n$. Let $\mathcal{Y} = \{\mathbf{Y}_1, \dots, \mathbf{Y}_n\}$.
- ▶ Each $\mathbf{Y}_i$ is Gaussian (given Σ),

$$\mathbf{Y}_i | \Sigma \sim \text{Normal}_p(\boldsymbol{\mu}, \Sigma),$$

where $\boldsymbol{\mu}$ is known.

- ▶ The likelihood is then

$$\prod_{i=1}^n f(\mathbf{y}_i | \Sigma) = |2\pi\Sigma|^{-n/2} \exp \left[-\frac{1}{2} \sum_{i=1}^n (\mathbf{y}_i - \boldsymbol{\mu})' \Sigma^{-1} (\mathbf{y}_i - \boldsymbol{\mu}) \right].$$

Bayesian analysis - Prior

- ▶ The parameter is neither scalar nor vector and we need to assign a prior for a random matrix Σ that is PD.
- ▶ Here, a natural prior is $\text{Wishart}_P(\mathbf{W}, \nu)$ where $\nu > P - 1$ and $\mathbf{W}$ is PD.
- ▶ However, the math is easier if we pick a Wishart prior for the inverse covariance (precision) matrix Σ^{-1} .
- ▶ If $\Sigma^{-1} \sim \text{Wishart}_P(\mathbf{W}, \nu)$, then $\Sigma \sim \text{Inverse-Wishart}_P(\mathbf{V}, \nu)$, where $\mathbf{V} = \mathbf{W}^{-1}$.
- ▶ This is the definition of the inverse-Wishart distribution.
- ▶ The inverse-Wishart prior for Σ has PDF

$$f(\Sigma) = \frac{|\mathbf{V}|^{\nu/2}}{2^{\nu P/2} \Gamma_P(\nu/2)} |\Sigma|^{-(\nu+P+1)/2} \exp \left[-\frac{1}{2} \text{Tr}(\mathbf{V}\Sigma^{-1}) \right].$$

Derivation of the posterior

- The posterior is

$$\begin{aligned} p(\Sigma|\mathcal{Y}) &\propto |\Sigma|^{-n/2} \exp \left[-\frac{1}{2} \sum_{i=1}^n (\mathbf{Y}_i - \boldsymbol{\mu})' \Sigma^{-1} (\mathbf{Y}_i - \boldsymbol{\mu}) \right] \\ &\quad \times |\Sigma|^{-(\nu+P+1)/2} \exp \left[-\frac{1}{2} \text{Tr}(\mathbf{V} \Sigma^{-1}) \right] \\ &= |\Sigma|^{-(n+\nu+P+1)/2} \exp \left[-\frac{1}{2} \text{Tr} \left(\left\{ \sum_{i=1}^n (\mathbf{Y}_i - \boldsymbol{\mu})(\mathbf{Y}_i - \boldsymbol{\mu})' + \mathbf{V} \right\} \Sigma^{-1} \right) \right]. \end{aligned}$$

- The posterior is

$$\Sigma|\mathcal{Y} \sim \text{Inverse-Wishart}_P(SSE + \mathbf{V}, n + \nu),$$

where $SSE = \sum_{i=1}^n (\mathbf{Y}_i - \boldsymbol{\mu})(\mathbf{Y}_i - \boldsymbol{\mu})'$.

Shrinkage

- ▶ The mean of an Inverse-Wishart $_P(\mathbf{V}, \nu)$ distribution only exists if $\nu > P + 1$.
- ▶ The prior mean (if it exists) is $\frac{1}{\nu - P - 1} \mathbf{V}$.
- ▶ The posterior mean of Σ is

$$\frac{SSE + \mathbf{V}}{n + \nu - P - 1}.$$

- ▶ It is common to take ν and all the elements of $\mathbf{V}$ to be small to give a non-informative prior.
- ▶ For large n and comparatively negligible ν and the elements of $\mathbf{V}$, the posterior mean is approximately same as the sample variance SSE/n .

Conjugate prior for a normal precision matrix

- ▶ The precision matrix is the inverse covariance matrix, $\Omega = \Sigma^{-1}$.
- ▶ If $\mathbf{Y}_i$'s have mean vector μ and precision matrix Ω , the likelihood is proportional to

$$\prod_{i=1}^n f(\mathbf{y}_i | \Omega) \propto |\Omega|^{n/2} \exp \left[-\frac{1}{2} \sum_{i=1}^n (\mathbf{y}_i - \mu)' \Omega (\mathbf{y}_i - \mu) \right].$$

- ▶ If $\Omega \sim \text{Wishart}_P(\mathbf{W}, \nu)$, then its PDF is given by

$$f(\Omega) = \frac{|\mathbf{W}|^{-\nu/2}}{2^{\nu P/2} \Gamma_P(\nu/2)} |\Omega|^{(\nu-P-1)/2} \exp \left[-\frac{1}{2} \text{Tr}(\mathbf{W}^{-1} \Omega) \right].$$

- ▶ In that case, we have $\Omega | \mathcal{Y} \sim \text{Wishart}([SSE + \mathbf{W}^{-1}]^{-1}, n + \nu)$.
- ▶ This is the exact same analysis as the inverse-Wishart prior for covariance.

Thank you!